Aquarium After Hours

Whodunnit? · Graphing Inequalities Name: _____

It was the Ocean Gala at the Bayview Aquarium. When the lights dimmed for the jellyfish show, the glass case holding the **Pearl of the Deep Blue** went dark, and when they came back on it was empty!

Six people were inside the aquarium, each in a different place, and there are six clever ways to carry a pearl out. You are the lead detective: find **who** took it, **where** they were, and **how** they got it out.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Captain Reyes Dive Instructor	☐ Shark Tunnel	☐ Hidden in a fish bucket
☐ Dr. Kelp Marine Biologist	☐ Touch Pool	☐ Smuggled in a dive bag
☐ Ms. Nori Gift Shop Owner	☐ Jellyfish Gallery	☐ Tucked in a stroller
☐ Mr. Finley Feeding Specialist	☐ Penguin Cove	☐ Wrapped in a wetsuit
☐ Coral Chen Tour Guide	☐ Gift Shop	☐ Disguised as a snow globe
☐ Prof. Tide Visiting Scientist	☐ Dive Deck	☐ Stuffed in a plush octopus

The culprit is _____ in _____ using _____

Show Your Work

Aquarium After Hours · Show every step in the matching box. Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

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A **dashed** boundary line passes through (0, 1) and (1, 3). The region **above** the line is shaded.

A. $y > 2x + 1$	Captain Reyes was checking tanks on the Dive Deck. The equipment log has his initials.
B. $y < 2x + 1$	Captain Reyes was posing for a photo with the penguins in Penguin Cove. The camera clock says 9:04 p.m.
C. $y \geq 2x + 1$	Prof. Tide was measuring water samples in the Shark Tunnel. The lab notebook has a 9:02 p.m. entry.
D. $y > \frac{1}{2}x + 1$	Every plush octopus in the Gift Shop was unzipped and checked. Only stuffing inside.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

Aquarium After Hours

A **solid** boundary line passes through (0, 1) and (1, 3). The region **below** the line is shaded.

A. $y < 2x + 1$	Prof. Tide was showing a sea star to a family at the Touch Pool. They took a photo together.
B. $y \leq 2x + 1$	The fish buckets were all emptied and rinsed. No pearl inside.
C. $y \geq 2x + 1$	Dr. Kelp was checking penguin eggs in Penguin Cove. The nest chart has his handwriting.
D. $y \leq \frac{1}{2}x + 1$	The plush octopuses on the shelf are all still sealed in plastic. None held the Pearl.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

Aquarium After Hours

A **dashed** boundary line passes through (0, 1) and (1, −1). The region **below** the line is shaded.

A. $y > -2x + 1$	Ms. Nori was handing out penguin stickers in Penguin Cove. A kid still has one on his shirt.
B. $y < -2x + 1$	Dr. Kelp was showing kids a starfish at the Touch Pool.
C. $y \leq -2x + 1$	Prof. Tide was giving a short talk in the Jellyfish Gallery. The whole audience stayed seated.
D. $y < 2x + 1$	A staff member squeezed every plush octopus. All were soft and squishy all the way through.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

Aquarium After Hours

A **solid** boundary line passes through (0, 1) and (−1, 3). The region **above** the line is shaded.

A. $y > -2x + 1$	Prof. Tide was buying a postcard in the Gift Shop. The receipt is time-stamped 9:03 p.m.
B. $y \leq -2x + 1$	Mr. Finley was tossing fish to the penguins in Penguin Cove while a crowd laughed.
C. $y \geq 2x + 1$	The plush octopus bin was counted before and after the gala. The numbers match.
D. $y \geq -2x + 1$	Every dive bag was searched at the exit. None held the Pearl.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

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A **dashed** boundary line has slope 2 and y-intercept −1. The region **above** the line is shaded.

A. $y > 2x + 1$	The giant plush octopus by the door was weighed and found to be only fluff.
B. $y < 2x - 1$	Prof. Tide was helping unload gear on the Dive Deck. The deckhand's log lists his name.
C. $y \geq 2x - 1$	Coral Chen was leading a tour group through Penguin Cove. The group counted heads at 9:05 p.m.
D. $y > 2x - 1$	Ms. Nori was ringing up souvenirs in the Gift Shop. The receipts are time-stamped.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #6

Aquarium After Hours

A **solid** boundary line has slope −2 and y-intercept −1. The region **below** the line is shaded.

A. $y < -2x - 1$	A volunteer saw Prof. Tide feeding the crabs at the Touch Pool when the lights went out.
B. $y \geq -2x - 1$	The feeding schedule shows Mr. Finley in Penguin Cove from 8:55 to 9:15 p.m.
C. $y \leq -2x - 1$	The strollers were checked at the door. Only snacks and blankets.
D. $y \leq -2x + 1$	The seams on every plush octopus are factory-sewn and intact.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

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A **dashed** boundary line passes through (0, −1) and (2, 0). The point (0, 0) is in the shaded region.

A. $y > 2x - 1$	Dr. Kelp was taking the water temperature in Penguin Cove. The reading was logged at 9:00 p.m.
B. $y \geq \frac{1}{2}x - 1$	A guest bought a plush octopus at 9:20 p.m. and cut it open at the exit. Just stuffing.
C. $y < \frac{1}{2}x - 1$	Prof. Tide was pointing out a hammerhead to students inside the Shark Tunnel.
D. $y > \frac{1}{2}x - 1$	Mr. Finley was feeding the sharks in the Shark Tunnel while a tour group watched.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

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A **solid** boundary line passes through (0, −1) and (−2, −2). The shaded region is **below** the line.

A. $y < \frac{1}{2}x - 1$	The Gift Shop owner shook every plush octopus and heard nothing rattle inside.
B. $y \leq 2x - 1$	Captain Reyes was answering kids' questions at Penguin Cove. A school group backs him up.
C. $y \leq \frac{1}{2}x - 1$	The wetsuits on the rack are all dry and flat. Nothing was wrapped in one.
D. $y \geq \frac{1}{2}x - 1$	Prof. Tide was taking notes on moon jellies in the Jellyfish Gallery. His pen ran out of ink there.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

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A **dashed** boundary line passes through (0, 2) and (2, 1). The point (0, 0) is in the shaded region.

A. $y < -\frac{1}{2}x + 2$	Coral Chen was leading a tour through the Jellyfish Gallery.
B. $y \leq -\frac{1}{2}x + 2$	Coral Chen was counting penguins in Penguin Cove for her tour script. Every one was present.
C. $y > -\frac{1}{2}x + 2$	Prof. Tide was coiling a rope on the Dive Deck. A diver says he never left the rail.
D. $y < -2x + 2$	Security scanned the plush octopuses with a handheld wand and found nothing hidden.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

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A **solid** boundary line passes through (0, 2) and (−2, 3). The shaded region is **above** the line.

A. $y \geq -\frac{1}{2}x + 2$	The snow globes on the shelf are sealed and unopened. The Pearl isn't in any of them.
B. $y \geq \frac{1}{2}x + 2$	Ms. Nori was taking a break on the bench at Penguin Cove. The security camera caught her there.
C. $y > -\frac{1}{2}x + 2$	The plush octopus display was photographed at 8:50 and 9:30 p.m., and every octopus looks the same in both.
D. $y \leq -\frac{1}{2}x + 2$	Prof. Tide was signing a book for a fan in the Gift Shop. The fan still has the autograph.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.